

Trial Test 6

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Evaluate

$$(\sqrt{2}-\sqrt{5}) * (\sqrt{20}+\sqrt{8})$$

a) $11 + \sqrt{2}$

b) -6

c) 4

d) $15 + \sqrt{2}$

2) Rearrange to make x the subject:

$$y = 5 - \sqrt{x^2 + 11}$$

a) $\sqrt{y^2 - 10y + 14}$

b) $\sqrt{y^2 - 5y + 14}$

c) $\sqrt{y^2 - 10y + 16}$

d) $\sqrt{y^2 - 5y + 14}$

3) Find $5n+2m$

$$5^n * 125^{2n} = 5^{56}$$

$$3^m = \frac{1}{9}$$

a) 44

b) 36

c) 38

d) 48

4) The 3rd and 7th term of an arithmetic progression are -9 and 11 respectively.

What is the 15th term?

a) 67

b) 35

c) 53

d) 51

5) An electronic device makes a beep after every 60 sec. Another device makes a beep after 62 sec. They beeped together at 10 a.m. The time(in min) after which they will next make a beep together at earliest is

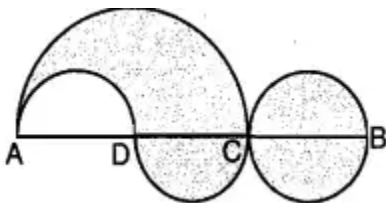
a) 10:53

b) 10:16

c) 10:31

d) 10:42

6) The given figure consists of three semi-circular arcs and one circle. The smaller arcs are equal. If $AB=6$ cm, $AC=4$ cm then what is the complete length of the two figures?



a) 6π

b) 5π

c) 4π

d) 8π

7) The line joining the point P (3,-4) to Q (Q,0) has a gradient of 2.

What is the value of q?

a) 8

b) 6

c) 5

d) 3

8) How many times do the function $f(x) = -x^4 + 2x^2 + 3$ intercept the x-axis?

a) 1

b) 2

c) 3

d) 0

9) In a cone, the diameter of a base is d. If the height is two times the diameter, what is the volume of this cone, in terms of d?

a) πd^2

b) $\frac{\pi d^2}{2}$

c) $\frac{\pi d^2}{6}$

d) $2\pi d^2$

10) A tank is $\frac{3}{5}$ filled with water. When 500 ml of water is poured out of the tank, the tank becomes $\frac{1}{2}$ full.

What is the capacity of the tank in litres?

a) 1.5

b) 5

c) 7.5

d) 3

11) A sphere is inscribed in a cylinder. Find the ratio of volume of sphere to volume of cylinder.

a) 3:4

b) 1:3

c) 1:2

d) 2:3

12) Convert 18 km/h to m/s

a) 5 m/s

b) 15 m/s

- c) 10 m/s
- d) 20 m/s

13) Find the number of diagonals in a regular polygon which has an external angle of 45 degrees.

- a) 20
- b) 22
- c) 30
- d) 24

14) For all numbers x and y the definition of the operation $\blacklozenge$ is $x\blacklozenge y = xy(x+4)$.

If $(-6)\blacklozenge a = -60$, then what is the value of a ?

- a) 0
- b) -1
- c) 3
- d) -5

15) One type of liquid contains 20% water and the second type of liquid contains 35% water. A glass filled with 8 parts of the first liquid and 5 parts of the second liquid. The water percentage in the new mixture is

- a) 25.56
- b) 25.67

c) 25.76

d) 25.75

16) What is the area of the largest square, in terms of r , that can be inscribed in a semicircle of radius ' r ' units?

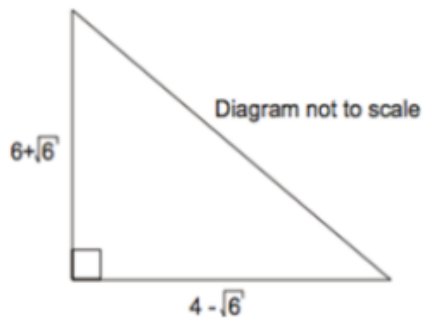
a) $\frac{4}{5}r^2$

b) r^2

c) $\frac{3}{5}r^2$

d) $\frac{2}{5}r^2$

17) A right-angled triangle is shown. Calculate the area of the triangle.



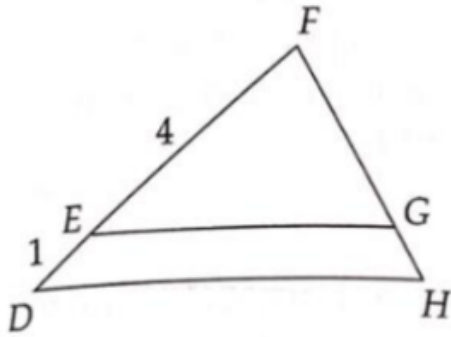
a) $9 - \sqrt{6}$

b) $9 - \sqrt{3}$

c) $15 + 5\sqrt{6}$

d) $9 + \sqrt{6}$

18) EG is parallel to DH, and the lengths of segments DE and EF are as marked. If the area of EGF is a , what is the area of DFH in terms of a ?



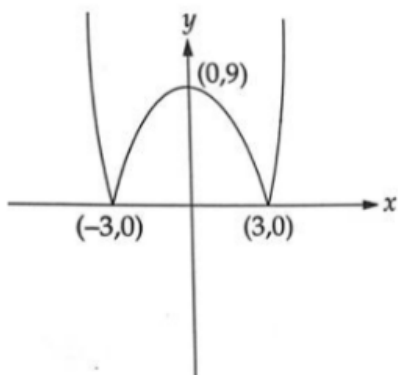
a) $\frac{16a}{25}$

b) $\frac{16a}{20}$

c) $\frac{5a}{4}$

d) $\frac{25a}{16}$

19) The graph could be the graph of which of the following functions?



a) $(x - 9)^2$

b) $9 - x^2$

c) $f(x) = -|x^2 + 9|$

d) $f(x) = |-x^2 + 9|$

20) Ten years ago, Ron's age was twice that of Hermione. Today, the ratio of their ages is 7:4.

What is the sum of their ages today?

a) 90

b) 130

c) 110

d) 120

21) Calculate

$$\frac{3.246 \times 10^6 + 3.246 \times 10^4}{1.623 \times 10^{11}}$$

Give your answer as an ordinary number, correct to 3 significant figures.

a) 0.0000102

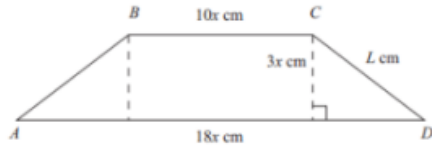
b) 0.0000202

c) 0.0000302

d) 0.0000402

22) The figure shows a symmetrical trapezium with $AD=18x$ cm, $BC=10x$ cm and $BA=CD=L$ cm. The perpendicular distance between AD and BC is $3x$ cm.

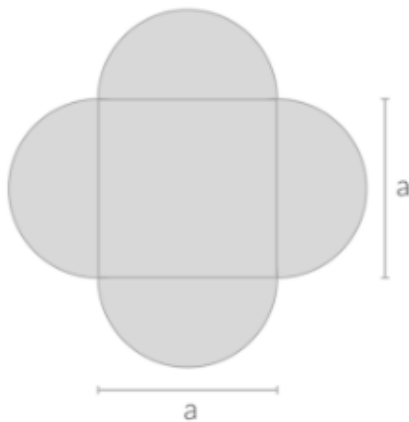
Find an expression, in terms of x , for the area of $ABCD$.



- a) $84x^2$
- b) $10.5x^2$
- c) $21x^2$
- d) $42x^2$

23) Consider the shape below.

Which expression best represents the shaded area?

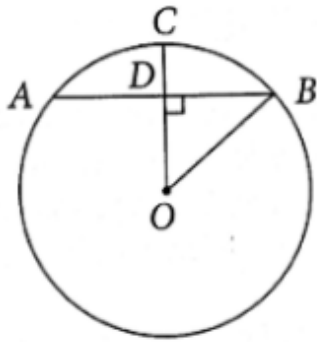


- a) $a^2 + \frac{1}{4}\pi a^2$
- b) $\frac{3}{2}a^2$

c) $a^2 + \frac{1}{2}\pi a^2$

d) $\frac{5}{2}a^2$

24) In the figure, O is the center of the circle. If AB has a length of 16 and OB has a length of 10, what is the length of CD?



a) $2\sqrt{3}$

b) 4

c) 2

d) $4\sqrt{3}$

25) The vertices of a rectangle have coordinates:

P (4,5)

Q (4,8)

R (10,8)

S (10,5)

PQRS is transformed by a clockwise rotation of 90° about P followed by a reflection in the x -axis.

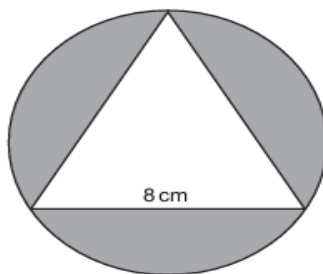
What are the coordinates of the final position of R?

- a) $(-8, -10)$
- b) $(-1, 11)$
- c) $(4, -1)$
- d) $(7, 1)$

26) An equilateral triangle of side 8 cm is drawn so that its vertices lie on the circumference of a circle, as shown in the diagram.

What is the total of the three areas shaded in the diagram, in cm squared?

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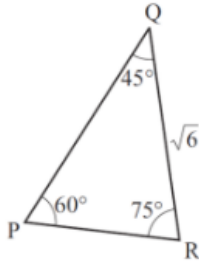


- a) $48(4\pi - \sqrt{3})$
- b) $\frac{16}{3}(4\pi - 6)$
- c) $24(\pi - \sqrt{3})$
- d) $\frac{16}{3}(4\pi - 3\sqrt{3})$

27) In triangle PQR

Angle QPR is 60 degrees, PQR is 45 degrees, QRP is 75 degrees, RQ is $\sqrt{6}$ units.

Find the length of side PR.



- a) 2
- b) $\sqrt{2}$
- c) $\sqrt{6}$
- d) $\sqrt{3}$

28) Two variable are connected by the relation: $A \sim \frac{1}{B^2}$. B is increased by 40%.

Describe the change in A in percentage terms.

- a) 29% decrease
- b) 51% decrease
- c) 49% decrease
- d) 44% decrease

29) Nicky and Christina are running a race, Since Christina is faster than Nicky, she gives him a 36-meter head start. If Christina runs at a pace of 5 m/s and Nicky runs at a pace of only 3 m/s, how many seconds Nicky have to run before Christina catches up to him?

a) 15 s

b) 18 s

c) 25 s

d) 12 s

30) Write the answer in fractional form.

$$0,3(6) + 0,(64)$$

a) $\frac{1003}{990}$

b) $\frac{1605}{990}$

c) $\frac{997}{990}$

d) $\frac{1317}{990}$